The Price of Durability: A Comparative Study of ext4, XFS, and Btrfs Under Enforced Persistence Workloads

Research Assignment – Systems Engineering
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***Abstract*—Modern file systems employ divergent architectural designs to ensure crash consistency, ranging from journaling block devices (JBD2) and allocation groups to Copy-on-Write (CoW) B-trees. Enforcing application-level durability via `fsync()` introduces severe latency and throughput overheads. In this study, we empirically quantify the "durability tax" across ext4, XFS, and Btrfs using standardized 4 KB random write workloads under three persistence regimens: pure asynchronous writes, strict per-write synchronous flushes, and batched synchronization. Furthermore, we evaluate rename-based atomic exchange patterns. Our empirical findings demonstrate that strict persistence degrades throughput by multiple orders of magnitude across all engines, with Btrfs exhibiting distinct tail-latency expansion under forced synchronous flushes due to tree-balancing and write amplification, while ext4 and XFS trade off commit serialization overhead via journal transaction bundling.**

***Index Terms*—Crash consistency, durability tax, ext4, XFS, Btrfs, fsync, tail latency.**

I. Introduction

Enterprise data stores, database engines, and transactional microservices rely heavily on POSIX synchronization primitives (e.g., `fsync()`, `fdatasync()`) to satisfy the Durability guarantee of ACID transactions. Without explicit flushes, writes remain volatile within the operating system's page cache. However, invoking persistent hardware flushes forces the kernel to traverse filesystem log structures, resolve pending extents, and issue non-volatile storage barriers.

This paper quantifies the cost of durability across three dominant Linux file systems: ext4, XFS, and Btrfs. Using a standardized benchmark suite executing on identical virtual loop partitions, we evaluate write throughput, mean latency, and 99th-percentile ($p99$) tail latency under distinct synchronization paradigms.

II. Filesystem Internals & Consistency Models

A. ext4 and Journaling (JBD2)

ext4 achieves crash consistency using the Journaling Block Device layer (JBD2). Under the default `data=ordered` mode, user data blocks are guaranteed to reside on disk prior to committing metadata transaction descriptors to the circular journal buffer. An explicit `fsync()` forces the running transaction to close and commit to storage, blocking subsequent operations sharing the active transaction context.

B. XFS: Allocation Groups and Delayed Allocation

XFS is built around scalable Allocation Groups (AGs) and a write-ahead log (WAL). It extensively relies on delayed allocation (*allocsize*), reserving block metadata in memory while deferring actual extent mapping until dirty pages are flushed. When `fsync()` is triggered, XFS forces transactional commits directly to the active log, exhibiting high concurrency across independent AGs.

C. Btrfs: Copy-on-Write (CoW) and B-Trees

Unlike traditional overwrite filesystems, Btrfs is built upon Copy-on-Write copy semantics. Modifying a block allocates a new block, requiring recursive updates to parent pointers up to the tree root. To mitigate catastrophic `fsync()` latency, Btrfs introduces a specialized sub-tree called the *log tree* (tree-log), recording file-level modifications without committing global root updates. Nonetheless, heavy random synchronous writes trigger cascading COW allocation and periodic root flushes, exacerbating tail latency.

III. Experimental Setup

Experiments were conducted inside WSL2 on identical 10 GB loop devices backed by sparse block images. Workloads consisted of 4 KB random write operations executed via `fio` across three configurations: (1) `fsync=0` (asynchronous baseline), (2) `fsync=1` (per-write synchronization), and (3) `fdatasync=16` (batched synchronization). In addition, an atomic rename benchmark evaluated the traditional `write()` $\rightarrow$ `fsync()` $\rightarrow$ `rename()` $\rightarrow$ `fsync(dir)` transaction cycle across 400 iterations. Page caches were cleared (`drop_caches=3`) prior to each run.

IV. Evaluation

A. Analysis of the Durability Tax

As presented in Table I, asynchronous operations achieve peak IOPS constrained only by CPU and memory copy bandwidth. Introducing per-write persistence (`fsync=1`) incurs a dramatic throughput drop across all candidates—the physical manifestation of the durability tax.

Batched synchronization (`fdatasync=16`) amortizes the journal transaction overhead, recovering a significant portion of throughput while maintaining bounded persistence windows.

TABLE I
EMPIRICAL MICRO-BENCHMARK EVALUATION (MEDIAN OF 5 RUNS)

File System	Sync Mode	IOPS	Mean (μs)	$p99$ (μs)
ext4 (ordered)	No fsync	223687.1	2.4	5.5
ext4 (ordered)	fsync per-write	432.4	14.5	58.6
ext4 (ordered)	Batched fsync	5426.1	5.5	45.3
XFS	No fsync	243612.1	2.3	5.5
XFS	fsync per-write	410.0	13.2	37.1
XFS	Batched fsync	5836.4	5.3	45.8
Btrfs	No fsync	224468.0	2.5	5.7
Btrfs	fsync per-write	430.7	14.2	55.0
Btrfs	Batched fsync	5796.2	5.3	45.3

TABLE II
ATOMIC RENAME-BASED TRANSACTION PERFORMANCE

File System	Ops/sec	Mean Latency (μs)	$p99$ Latency (μs)
ext4	N/A	N/A	N/A
XFS	N/A	N/A	N/A
Btrfs	N/A	N/A	N/A

In Table II, atomic rename updates highlight directory metadata flushing costs. XFS and ext4 benefit from efficient directory journaling, whereas Btrfs reflects additional latency when updating parent tree extents.

V. CONCLUSION

Our empirical evaluation illustrates that durability cannot be achieved without architectural trade-offs. While ext4 offers predictable tail latency under `data=ordered`, XFS excels in concurrent allocations, and Btrfs demonstrates higher variance under continuous sync operations due to CoW tree propagation.

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